



FADO

Meeting 2

Tiago Silva

CISUC · Data-Centric AI Lab

Department of Informatics Engineering, University of Coimbra



Outline

1. **Weekly reviews:** KANs;
2. **Review pass:** what we found, what we fixed, a correction;
3. **Decisions taken;**
4. **The λ controller:** the answer to the novelty critique;
5. **First results** and how they happened;
6. **Ablation plan**, open questions and next steps.

KANs

KAN: Kolmogorov–Arnold Networks

Ziming Liu^{1,4*} Yixuan Wang² Sachin Vaidya¹ Fabian Ruehle^{3,4}
James Halverson^{3,4} Marin Soljačić^{1,4} Thomas Y. Hou² Max Tegmark^{1,4}

¹ Massachusetts Institute of Technology

² California Institute of Technology

³ Northeastern University

⁴ The NSF Institute for Artificial Intelligence and Fundamental Interactions

Kolmogorov-Arnold Networks

Activation functions are placed on edges instead of nodes.

D^3 KAN: Dual Denoising and Data-Complexity-Adaptive Kolmogorov-Arnold Networks for Long-Term Time Series Forecasting

Minrui Wang^{1,2}, Jiwei Qin^{1,2*}, Di Wang^{3,4}, Dacheng Wang⁵,
Zhiqi Shen^{3,4}

D^3 KAN

A denoising and data-complexity-adaptive KAN for long-term forecasting.

The Fortieth AAAI Conference on Artificial Intelligence (AAAI-26)

Learning Fair Representations with Kolmogorov-Arnold Networks

Amisha Priyadarshini¹, Sergio Gago-Masague¹

¹Department of Computer Science, University of California, Irvine, USA
apriyad1@uci.edu, sgagomas@uci.edu

Learning Fair Representations with KANs

KANs applied to learning fair representations.

We change the objective function from $\mathcal{L} + \lambda \mathcal{P}(\theta)$ to $\min_{\theta} \mathcal{L}(\theta)$ s.t. $DP(\theta) \leq \epsilon$ so we can ensure more flexibility during drift and avoid the need for a per dataset tunable hyperparameter λ

Theorem 1

$\min_{\theta} \mathcal{L}(\theta)$ s.t. $DP(\theta) \leq \epsilon$ is equivalent to $\min_{\theta} \max_{\mu \geq 0} \mathcal{L}$

Proof.

Let $DP = \max_g |D_g|$, $D_g = \hat{r} - r_g$ where r_g is group g 's positive-prediction rate and $\hat{r}$ is the mean of those rates (e.g. with two groups deviations are always mirror images meaning $D_0 = -D_1$). Using the fact that $|x| \leq \epsilon \equiv -x \leq \epsilon \wedge x \leq \epsilon$ then we can rewrite $|D_g| \leq \epsilon$ as $D_g \leq \epsilon$ and $D_g \geq -\epsilon$: Same requirement without absolute value and, additionally, we know which side is being violated.

We can now rewrite the Loss function using the Lagrange multiplier μ as follows: $\mathcal{L}(\theta, \mu) = \mathcal{L}(\theta) + \sum_g (\mu_{g+}(D_g - \epsilon) + \mu_{g-}(-D_g - \epsilon))$ with $\mu_k \geq 0, \forall \mu_k$.

If we assume a fair classifier where $DP - \epsilon \leq 0$ then $\mu(DP - \epsilon) \rightarrow -\infty$ as μ increases, whereas if we assume an unfair classifier where $DP - \epsilon > 0$ then $\mu(DP - \epsilon) \rightarrow +\infty$.

Therefore, the Lagrange multiplier μ will be maximized when the classifier is unfair and minimized when the classifier is fair. This means that the optimization problem can be reformulated as a min-max problem, where we minimize the loss function with respect to the model parameters θ while maximizing it with respect to the Lagrange multiplier μ (if we instead chose to minimize the μ player would just choose 0 everytime). This leads to the equivalent formulation:

$$\min_{\theta} \max_{\mu \geq 0} \mathcal{L}(\theta, \mu).$$

□

This also means we can find the price, μ , by dual ascent making λ adaptative, dataset agnostic and not requiring hyperparameter tuning. This is a more flexible approach that allows the model to adapt to changing conditions and ensure fairness without the need for manual tuning of λ . The dual ascent finds the optimal value of μ in the following way:

$$\frac{\partial \mathcal{L}}{\partial \mu_{g\pm}} = \pm(D_g - \epsilon).$$

Updating μ as: $\mu_{g\pm} = \text{clip}(\mu_{g\pm} + \eta(\pm D_g - \epsilon), 0, \lambda_{\max})$ where η is the learning rate and $\lambda_{\max}$ is the maximum value of λ (which we can set to infinity). This way we keep non-negative prices and bound the duality requirement + accuracy trade-off.

How can we estimate D_g ?

We see one person at a time $(a_t, \hat{y}_t)$, which is not a meaningful rate for every group - we need to be right on average. The Horvitz-Thompson estimator states:

Let $Y_i, i = 1, \dots, n$ be an independent sample from n of $N \geq n$ distinct stratum with mean μ . Suppose π_i is the inclusion probability (i.e. sample belongs to the i -th stratum) then: $\hat{Y}_{HT} = \sum_{i=1}^n \frac{Y_i}{\pi_i}$ and mean estimate $\hat{\mu}_{HT} = \frac{1}{N} \sum_{i=1}^n \frac{Y_i}{\pi_i}$ is unbiased for μ .

In our case, we want to estimate r_g (equal it on average) and we can use the Horvitz-Thompson estimator to do so. We can estimate r_g as follows:

$$u_{t,g} = \frac{\hat{y}_{t=1}}{\pi_g} \text{ with probability } \pi_g \times r_g \text{ otherwise } 0.$$

On average:

$$\pi_g \times r_g \times u_{t,g} = r_g$$

Theorem 2

$\mathbb{E}[v] = D_g$: *The estimator is right on average.*

Proof.

Let the estimator $v_{t,g}$ be defined as $\frac{1}{G} \sum_h (u_{t,h} - u_{t,g})$. Because $\mathbb{E}[u_{t,g}] = \frac{1}{\pi_g} (\pi_g \times r_g) + 0(1 - \pi_g \times r_g) = r_g$, we have:

$$\mathbb{E}[v_{t,g}] = \frac{1}{G} \sum_h (\mathbb{E}[u_{t,h}] - \mathbb{E}[u_{t,g}]) = \frac{1}{G} \sum_h (r_h - r_g) = \hat{r} - r_g = D_g.$$

And thus we prove that the estimator is right on average. □

Limitations: Samples are independent given the model but not identically distributed (in online learning model's distribution changes over time)